DBAS32100 group project Template

The project for this course is worth 25% of your mark and

Introduction

In this project you will assist the fictional organization Pet Rescue Charity with managing their donations data. The charity organizes an annual donation drive across the city. The city is divided into donation areas. Every donation area consists of approximately 6 postal codes. A group of volunteers and one volunteer leader is assigned to each area. The volunteers go door-to-door in their designated areas and collect donations from the residents (the donors). The donations can be in the form of cash, cheque or credit card.

At the end of every week each volunteer leader will fill in a list showing the donations collected in their area and send it to the charity main office. The staff in the main office will then run a process that you will design to load the list to a Central Donation Repository. The process will reject non-valid entries. You will also create a data mart which will have data loaded from the Central Donation Repository. This data mart will be used for reports and to perform analysis based on interviews that have been conducted with Pet Rescue Charity.

Data Sources

Central donations repository

The Central Donations Repository is the main database where all the information on donations is stored it consists of Donation, Address, and volunteer tables

Your group will create the Central Donation Repository tables on your Project Oracle database instance. Below are the table creation statements. Every group will be provided the connection information for their project database. you will connect to the database as user name ‘prc’ in order to create these tables

1. Donation table

Every row in this table is a donation made by a donor. Below is the script that creates donation table

```
CREATE TABLE Donation
(Don_ID NUMBER PRIMARY KEY,
Donor_First_name VARCHAR2(16),
Donor_Last_name VARCHAR2(16),
Donation_Date Date NOT NULL,
Donation_Amount NUMBER(7,1) NOT NULL,
Address_ID  NOT NULL,
Volunteer_ID NOT NULL,
CONSTRAINT fk_don_add
FOREIGN KEY (Address_ID) REFERENCES Address(Address_ID),
CONSTRAINT fk_don_vol
FOREIGN KEY (Volunteer_ID) REFERENCES Volunteer(Volunteer_ID));
```

2. Address Table

This is an address lookup table. It has a list of all the known addresses

```
CREATE TABLE Address(
Address_ID NUMBER PRIMARY KEY,
Unit_Num VARCHAR2(6),
Street_Number NUMBER NOT NULL,
Street_Name VARCHAR2(24) NOT NULL,
Street_Type VARCHAR2(12) NOT NULL,
Street_Direction CHAR(1),
Postal_Code CHAR(7) NOT NULL,
City VARCHAR2(16) NOT NULL,
Province CHAR(2) NOT NULL,
CONSTRAINT St_Dir CHECK
(Street_Direction IN ('E', 'W', 'N', 'S')));
```

3. Volunteer Table

This table has information on the volunteers. Consider yourself to be the volunteers so you need to insert your information in this table

```
CREATE TABLE Volunteer(
Volunteer_ID NUMBER PRIMARY KEY,
First_name VARCHAR2(16),
Last_name VARCHAR2(16),
Group_Leader,
CONSTRAINT fk_vol_lead
FOREIGN KEY (Group_Leader)
REFERENCES Volunteer(Volunteer_ID)
);
```

Your Oracle instance has the following schemas already created for your group to use:

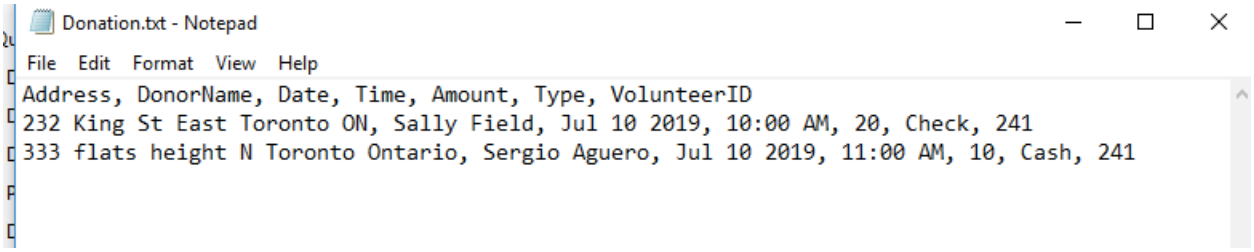
- PRC - is used to store donations, donor addresses and volunteer information in the Central Donation Repository you will use PCR username in order to create the above three tables.
- Datamart - is used to store the data mart star schema you will use Datamart user in order to create the star schema.

Users - the last two users (marked with *) must be created by you as task 6 describes later:

- Users SYS and SYSTEM are for database administration.
- PRC is the owner of the tables (will create the tables)
- Datamart - Is the owner of the datamart tables (will create the tables).

Donation List

This is the list that the volunteer group lead sends to the main office. It is a comma separated text file of the following format. The below image is just a sample of how the file should be structured. You need to create at least 3 different csv files with 20 entries each to test your process.



Addresses Table

The address table that the charity maintains in their oracle database is not updated and it often gets out of date. However, there is another department in the organization has an address table that they regularly update and keep current. The table is in SQLServer database and below is the connection information to the server

Make sure that you have setup Sheridan VPN. You can not access the server unless you are connected to Sheridan vpn.

Server Name: dbr.fast.sheridanc.on.ca
Port: 1433
Database name: Integration
Schema Name: dbo
User Name: DataIntegrator
Password: Sher1dan

Tasks

Task 1 – write scripts or document a process that refreshes (copy new or changed addresses) the address table in the oracle database with the addresses in the master table (the table in SQL server database). You will need to transform some data types and generate a sequential id for the address. Also note that the ids of some addresses will be used as foreign key in donation table your solution should accommodate this fact. It does not have to be one scrip. You can make this task as a sequence of separate processes

Task 2 – You need to create a process that loads the donations list to the central donations table: In this task you will load the data from the csv file into the Donation table, You need to make sure that only the donations with valid addresses are inserted into the table. Donation records with erroneous addresses or incorrect volunteer ids must be rejected. Also make sure to reject the donations that have nulls for the mandatory columns in your database (Not null columns). Your process must generate csv files to the volunteer group leaders with the records that were rejected in their area.

You need to test your data by loading donations of 60 distinct donor names and addresses across at least 3 separate CSV files. This is required regardless of how many students are in your group. Each CSV file should have both valid and invalid entries, which is described later.

Task 3 – Create a star schema and ER diagram for the donations data mart. The grain of the schema should be the combination of date at the level of day, the address and the volunteer containing the number (count) of donations and the sum of the value of donations.

The date dimension should include the following fields: date (2021-07-01, 2021-07-02), year (2020, 2021), month_number (1...12), day_number_in_month (1...31), month_name_short (Jan, Feb…), month_name_long (January, February…)
The address dimension should include the fields: postal code, address
The volunteer dimension should include the fields: volunteer number, volunteer name

You need to connect to the database with datamart user when creating the star schema

Task 4 - Create a process to Load the data to the star schema from the central donation repository

Task 5 - Create views to address the following:

- The number of donations and total donations for each date showing the date hierarchy (year, long month name, day).

Date	Year	Month	Day	Number of Donations	Total Donation Amount
2021-04-28	2021	April	28	3	$150
2021-04-29	2021	April	29	5	$700
…					

- The number of donations, sums and average donations by location hierarchy (postal code and address)

Postal Code	Address	Number of Donations	Total Donation Amount	Average Donation
L6J 7H7	--	3	$330	$110
L6J 7H7	2970 Mulberry Dr. Oakville ON	1	$150	$150
L6J 7H7	2968 Mulberry Dr. Oakville ON	2	$70	$35
L6J 7H3	--	2	$80	$40
L6J 7H3	2938 Mulberry Dr. Oakville ON	1	$60	$60
L6J 7H3	2934 Mulberry Dr. Oakville ON	1	$20	$20
…				

- The number of donations, sums and average by volunteer leader and volunteer

Volunteer Leader	Volunteer	Number of Donations	Total Donation Amount	Average Donation
Jim Kirk	--	14	$640	$45.71
Jim Kirk	Leo McCoy	6	$240	$40
Jim Kirk	Monty Scott	8	$400	$50
Jean Picard	--	10	$600	$60
Jean Picard	Geo La Forge	4	$200	$50
Jean Picard	Dee Troi	6	$400	$66.67
…				

Task 6 Basic Security

- Create a user named DMLUser and give the user permissions to implement all DML on address, donation, and volunteer tables
- Create a user named Dashboard and give the user read permissions on the views

Deliverables

Report (Group Work) 75%

Create a word document that describes and illustrates the processes that you did. The document must include 6 distinct sections. Each of the section will be describing and illustrating one of the 6 tasks. You must describe the processes in details and include all the code that you used to create the process, you also need to state what software did you use for every task

It is important to note that I will be marking your tasks based on what you wrote in the report and it is your responsibility to make sure that your report reflect the work that you did. You will not get any marks for work that does not show in your report.

You also need to submit all the files that have your processes and codes (for example Tlend project or sql files) you also need to clearly identify for every the location of the artifacts (Tables, procedures, etc) in the Oracle database that you will be provided for the project. Failing to provide the location of the artifacts will result in 0 grade for the task

Presentation 25%

Students will present their projects to the class. Presentation time should be at most 5 minutes with 3 to 5 minutes of question time. Each student in the group will present the part of the project they worked on. The presentation must clearly explain the development process of every task and the difficulties the students face. Students who will not participate will get a 0 on the presentation grade. 5% of the grade will be on the clarity of the visual aid (e.g. PowerPoint), and 20 % will be graded individually based on the student's contribution and ability to explain their tasks.

Evaluation rubric for the project document (group work)

Item	**Criteria**			
Presentation	-Cover page with the students names -students names are on the header of every page of the report -The report is formatted in a clear manner -The report clearly describes the work **(5 %)**	One or two of the components described in the column to left are not present or done as prescribed **(3 - 4%)**	Three of the components described in the column to left are not present or done as prescribed **(1 - 2%)**	All of the components described in the column to left are not present or done as prescribe **(0%)**
Task 1	The addresses are correctly transferred. All the references in the address table in Oracle are preserved. The methodology to transfer addresses can be applied periodically **(10 - 8%)**	The addresses are correctly transferred. References are not preserved. The methodology to transfer addresses can be applied periodically **(7 - 6%)**	The addresses are correctly transferred. References are not preserved. The methodology to transfer addresses can not be realistically applied periodically **(5 - 1%)**	The addresses were not transferred **(0%)**
Task 2	Donations were loaded to oracle from text file. All the errors were rejected. The errors were extracted to a text file The methodology be applied	Donations were loaded to oracle from text file. All or some of the errors were not rejected. The errors were extracted to a text file The methodology can be applied periodically **(10-7%)**	Donations were loaded to oracle from text file. All or some of the errors were not rejected. The errors were not extracted to a text file The methodology can not be realistically	Donations were not loaded to oracle **(0%)**

	periodically **(15 - 11%)**		applied periodically **(6-1%)**	
Task 3	The ER diagram used the correct notations. The facts and dimensions tables are correct. The grain of the schema is correctly defined **10 - 8%)**	The ER diagram used the correct notations. The facts and dimensions tables are correct. The grain of the schema is not correctly defined **(7 - 6%)**	ER diagram does not use correct notation The facts and dimensions tables have errors. The grain of the schema is not correctly defined **(5 - 1%)**	The ER diagram is missing **(0%)**
Task 4	-Create table statements for the star schema are correct -The process or script to load data to the dimensions is correct -The process or script to load data to the facts is correct **(15 -11%)**	-Create table statements for the star schema are correct -The process or scripts to load data to the dimensions is correct -The process or scripts to load data to the facts is not correct **(10-7%)**	-Create table statements for the star schema are correct -The process or scripts to load data to the dimensions is not correct -The process or scripts to load data to the facts is not correct **(6-1%)**	The task is not done **(0%)**
Task 5	The three views are correctly created **(10 - 8%)**	Two of the three views are correctly created **(7 - 6%)**	One of the three views is correctly created **(5 - 1%)**	None of the views are correctly created **(0%)**
Task 6	-Users are created correctly -Correct privileges are applied **(10 - 8%)**	-Users are created correctly -Some privileges are not correct **(7 - 6%)**	-Some users are not correct -Some privileges are not correct **(5 - 1%)**	Users were not created **(0%)**
Total	80			

Presentation

Group grade on Presentations	-Clear presentation aid - well-organized **(5)**	- Moderate clarity and/or organization **(3)**	Limited to non-existing presentation aid **(0-1)**
Individual grade	More than equal contribution to other members And demonstrated leadership **(15-20)**	Equal contribution 9contributed to third of the project of there are 3 numbers) **(10)**	less than equal contribution **(5-0)**
• State the difficulties that you faced in the project	Explanation shows understanding of the topic, technologies used and clear idea on why the student used a certain technology **(10%)**	Explanation shows some understanding of the topic and technologies used **(5%)**	Explanation shows some understanding of the topic and technologies used **(0-2%)**

Due Dates

Group Report	**Week 13**
Presentation	**Week 13**